# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
## Department of Computer Science and Engineering
## ASSESSMENT TOOL 1 – INDUSTRY PROBLEM TASK (5 Questions)

| | | | |
|---|---|---|---|
| **Course Code:** | **DSA0605-slot c** | **Course Title:** | Data Handling and Visualization |
| **CO Assessed:** | CO2 – Precision (BL3) | **Assessment:** | Assessment Tool 1 – Industry Problem Task |
| **Weightage:** | 40% of CIA | **Total Marks:** | 50 (5 Questions × 10 Marks) |
| **CO2 Statement:** | *Apply appropriate visualization methods to represent distributions, proportions, and relationships in data.(BL3)* | | |
| **Student Name:** | G Kushwanth | **Reg. No.:** | 192424150 |
| **Section / Batch:** | 2024(AI&DS) | **Date:** | |

### Code of Conduct
*I G Kushwanth*(192424150) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student:

### Instructions
- This test contains 5 questions, each carrying 10 marks. Total: 50 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

| Section / Topic | Focus | Q Nos | Marks |
|---|---|---|---|
| **Categorical Data Visualization** | Comparing category-wise revenue, proportions, and seasonal trends using bar, grouped, and stacked charts | 1 | 10 |
| **Distribution Analysis** | Understanding defect distribution and variability using histograms, boxplots, and violin plots | 2 | 10 |
| **Relationship Visualization** | Exploring relationships between variables using scatter plots, heatmaps, and multivariate techniques | 3 | 10 |
| **Geospatial & Temporal Visualization** | Analyzing delivery performance using maps, time-series plots, and heatmaps | 4 | 10 |
| **Distribution Comparison** | Comparing patient data distributions using ridgeline plots, density plots, and boxplots | 5 | 10 |
| | | **GRAND TOTAL:** | **50 Marks** |

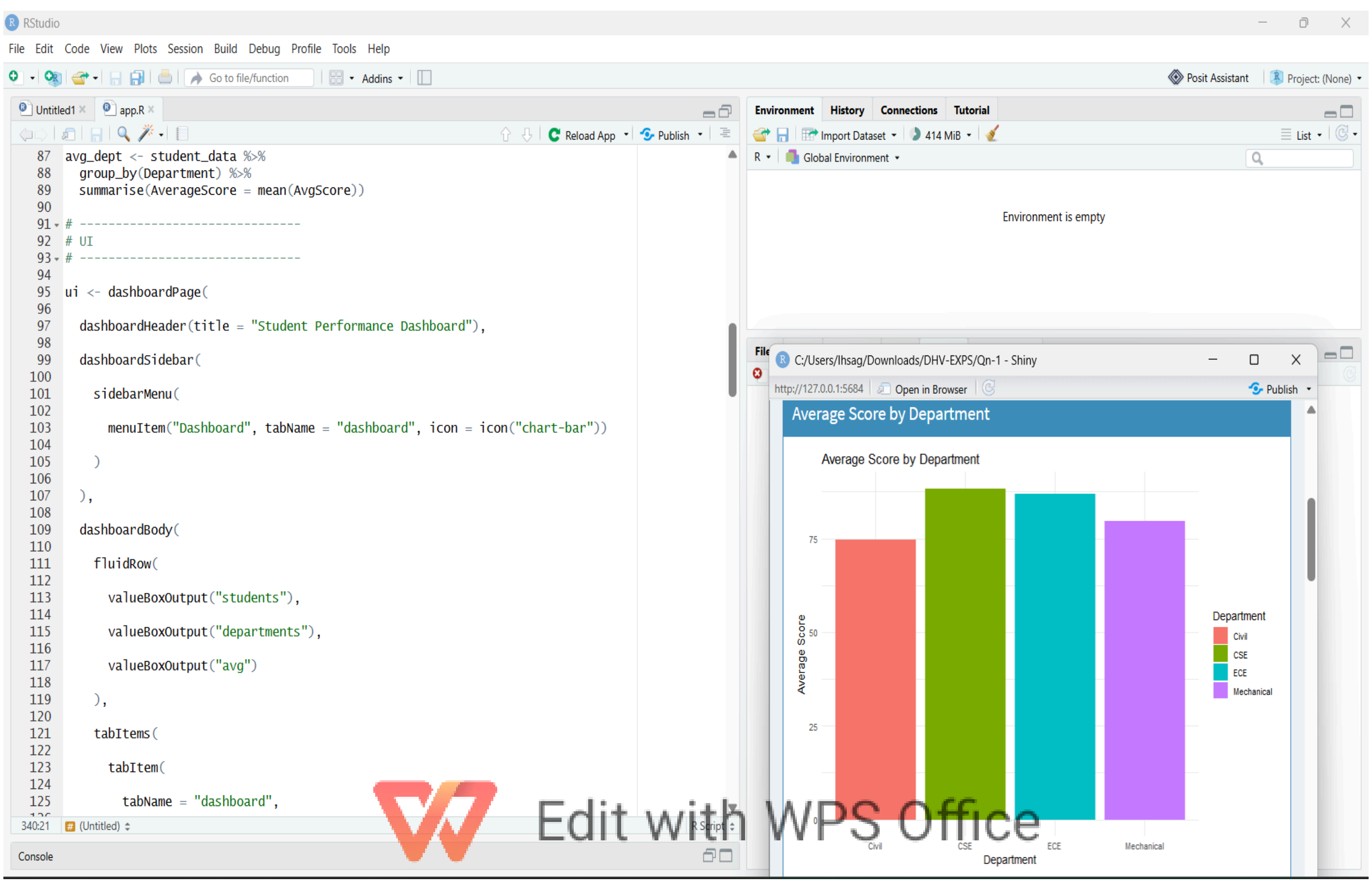
# Question 1: Academic Performance Analysis Across Departments

Problem: A university wants to compare average scores between departments (CS, Mechanical, Civil, Electronics), find the top-performing department, and see how much scores vary within each one.

## Recommended Visualizations

| Visualization | Purpose in this Scenario |
| --- | --- |
| Bar Plot | One bar per department showing its average score — easiest way to rank departments. |
| Grouped Bar Chart | Places bars for two or more things (e.g. Sem 1 vs Sem 2 average) next to each other for each department. |
| Histogram | Shows how many students scored in each mark range within one department. |
| Boxplot | Shows median, spread, and outlier students for each department, side by side. |

## Outputs



### Why These Charts Work

- **Bar plot:** Departments are categories, and we just need one number (average) per department — a bar chart makes ranking instant.
- **Grouped bars:** Useful when comparing more than one score per department without mixing up the bars.
- **Histogram:** An average can hide the real picture — a histogram shows if most students scored close to the average or if marks are scattered.
- **Boxplot:** Boxplots let us compare many departments at once and instantly spot which ones are consistent and which have unusual (outlier) students.

### How This Helps the University

- Quickly find departments that need extra academic support.
- Track if performance is improving or falling across semesters.
- Spot departments where marks are very spread out, meaning some students need more attention.
- Identify individual outlier students early for counselling.

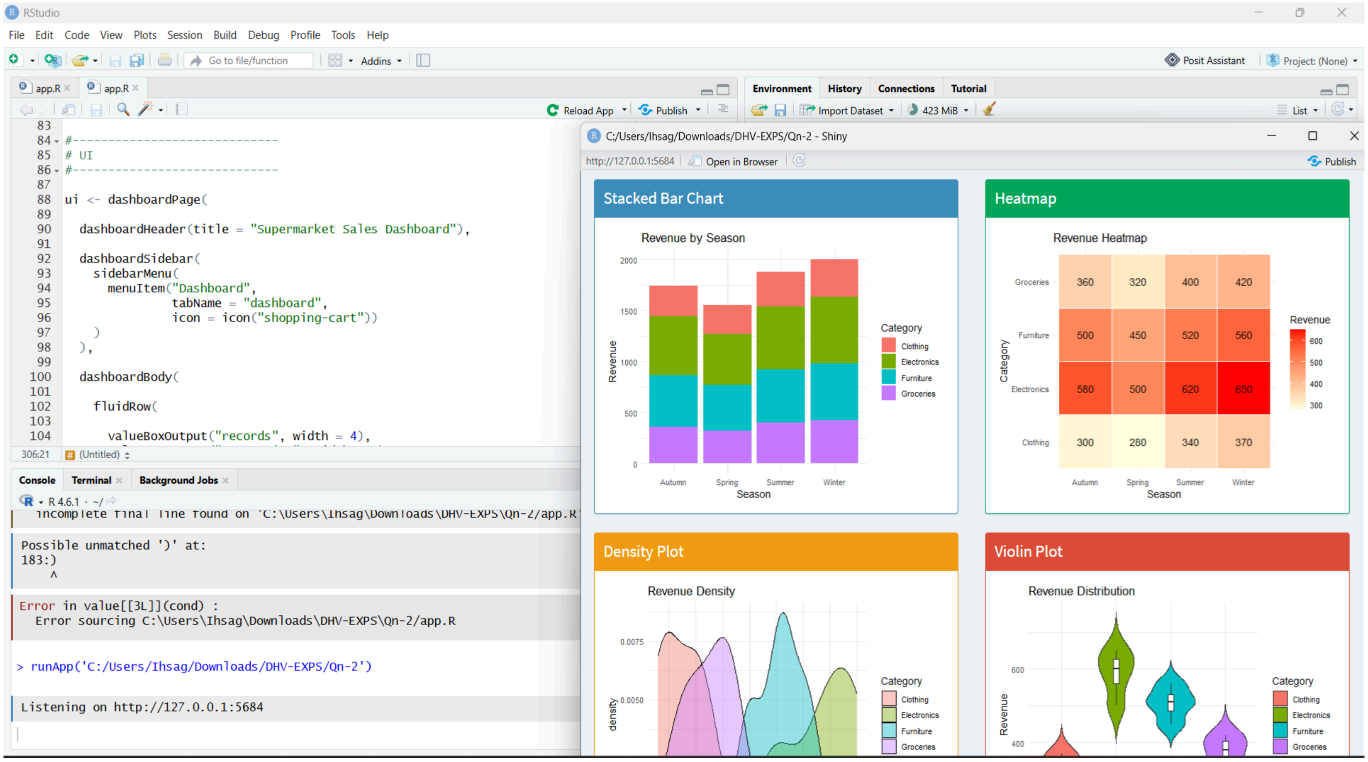
## Question 2: Supermarket Customer Purchase Analysis

Problem: A supermarket wants to see how much each product category sells in each season, and which categories bring in the most revenue.

### Recommended Visualizations

| Visualization | Purpose in this Scenario |
| --- | --- |
| Stacked Bar Chart | One bar per season, split into colored sections for each category — shows total sales and category share together. |
| Heatmap | A grid of category vs season with color showing sales amount — bright cells = high sales. |
| Density Plot | A smooth curve showing how purchase amounts are spread for a group of customers. |
| Violin Plot | Like a boxplot but also shows the shape of spending, useful for comparing categories. |

### Output




## Why These Charts Work (Simple Justification)

- **Stacked bars:** Shows the total sales for a season and how much each category contributed, in one chart.
- **Heatmap:** With many categories and seasons, color is faster to scan than a table of numbers.
- **Density plot:** Shows the shape of customer spending clearly, without needing to pick bin sizes like a histogram.
- **Violin plot:** Reveals patterns a plain average would hide — for example, many small buyers plus a few bulk buyers.

## How This Helps the Business

- Plan stock and staff levels ahead of high-sales seasons.
- Spot low-performing category-season combinations for discounts or promotions.

- Understand different customer spending habits for targeted offers.
- Design better bundle deals based on real buying patterns.

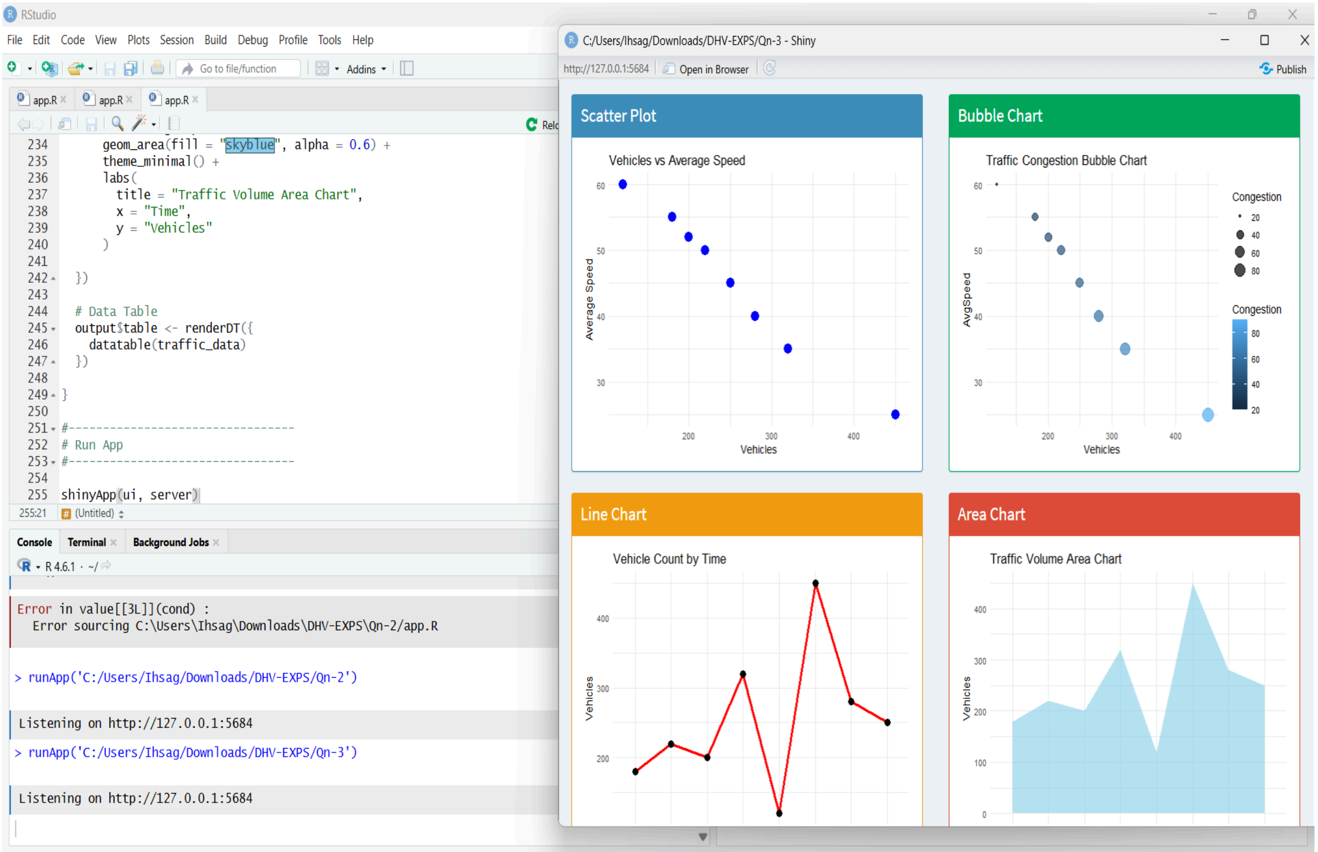
## Question 3: Smart City Traffic and Transportation Analysis

Problem: A city wants to find traffic hotspots and understand how congestion, speed, and travel time change by location and time of day.

### Recommended Visualizations

| Visualization | Purpose in this Scenario |
|---|---|
| Geospatial Map | Shows congestion or speed directly on a city map — hotspots are visible at a glance. |
| Heatmap (Location × Time) | Grid of locations vs hours of the day, colored by congestion level. |
| Temporal (Line) Plot | Line chart of a metric like speed over the day, showing peaks and dips. |
| Multivariate Plot | Combines several variables (speed, volume, congestion) in one chart, e.g. a bubble chart. |

### Sample Charts




## Why These Charts Work (Simple Justification)

- **Geospatial map:** Traffic data is about places, so showing it on a real map is the most natural way to see it.
- **Heatmap:** Traffic follows daily patterns — this chart makes rush-hour bottlenecks obvious immediately.
- **Line plot:** Best way to see how a value changes over time and to check if it is getting better or worse.
- **Multivariate plot:** Congestion depends on several factors together, so one combined chart shows the fuller picture.

### How This Helps City Planning

- Decide where new roads, signals, or flyovers are needed most.
- Time traffic signals and police duty around real rush hours.
- Check whether a new policy actually reduced congestion over time.
- Understand which factors cause the worst jams, to plan better routes.

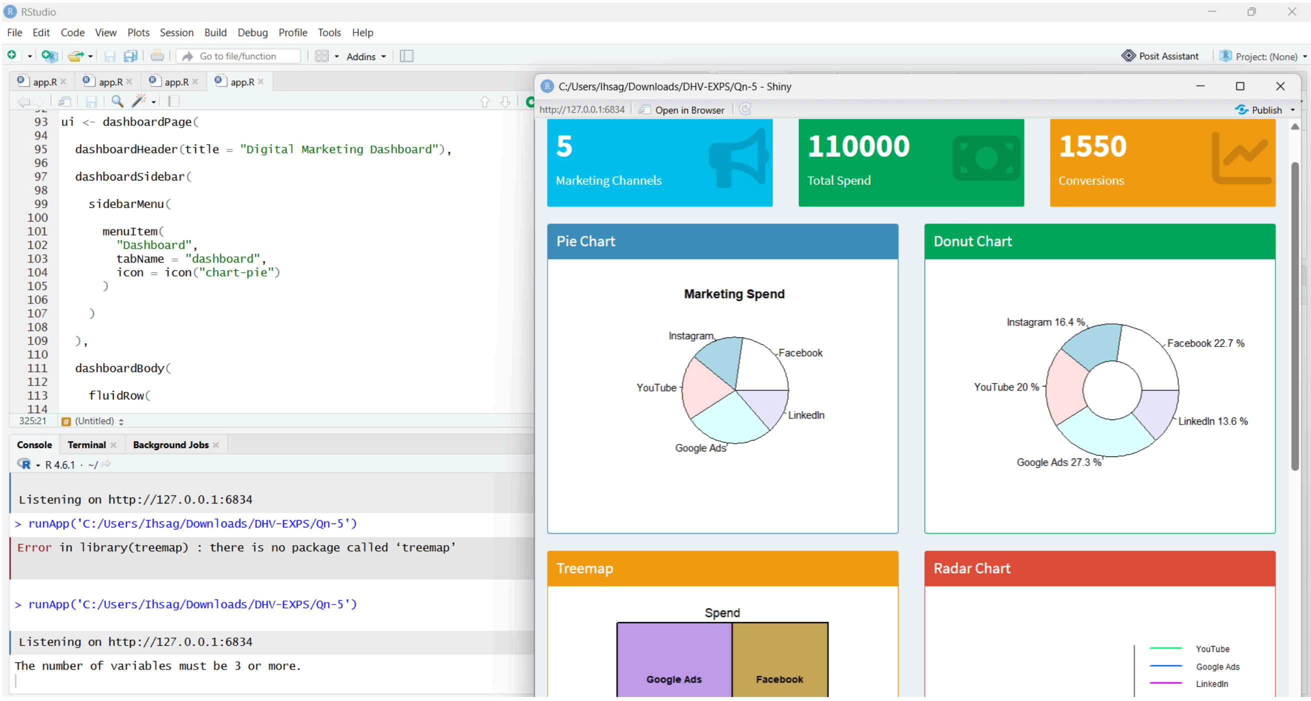
## Question 4: Digital Marketing Engagement Analysis

Problem: A marketing company wants to know how likes, shares, comments, watch time, and follower growth relate to each other and to campaign success.

### Recommended Visualizations

| Visualization | Purpose in this Scenario |
|---|---|
| Scatter Plot | Plots two metrics against each other (e.g. watch time vs shares) to see if they move together. |
| Heatmap (Correlation) | Grid showing how strongly every pair of metrics is related, using color. |
| Network Visualization | Shows how content spreads between accounts through shares and mentions. |
| Multivariate Plot | Bubble chart combining 3+ metrics at once, e.g. size of bubble = follower growth. |

### Outputs:




## Why These Charts Work (Simple Justification)

- **Scatter plot:** The simplest way to see if two metrics rise and fall together, and to spot unusual posts.
- **Correlation heatmap:** With many metrics, this shows all relationships at once instead of many separate charts.
- **Network graph:** Engagement spreads through people sharing content — a network chart shows this connection pattern that other charts cannot.
- **Multivariate plot:** Campaign success usually depends on more than one factor together, not just one metric alone.

## How This Helps Marketing Strategy

- Find which metric best predicts a successful campaign.
- Avoid tracking too many metrics that just repeat the same information.
- Identify key influencers driving the most reach.
- Design campaigns around the combination of factors that work best together.

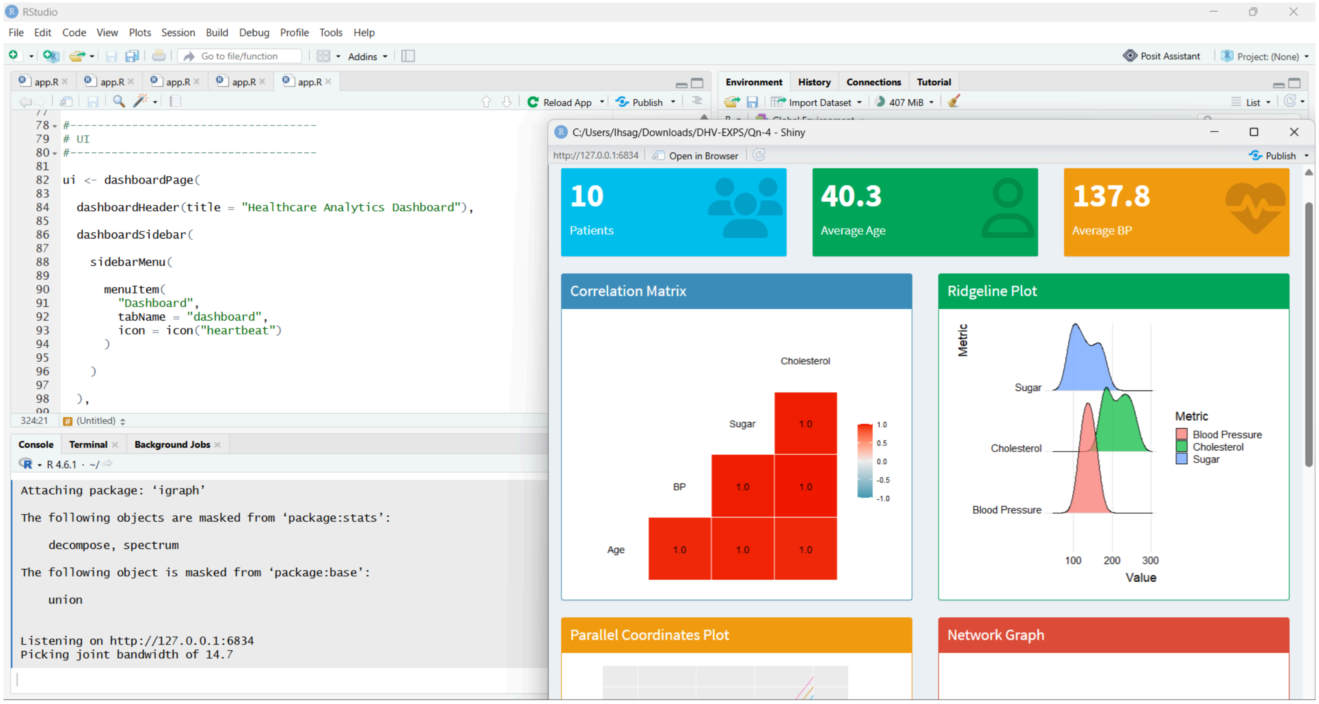
## Question 5: Healthcare Patient Recovery Analysis

Problem: A hospital wants to compare recovery rates across different patient groups and disease types, and understand overall healthcare trends.

### Recommended Visualizations

| Visualization | Purpose in this Scenario |
|---|---|
| Ridgeline Plot | Stacks recovery-rate curves for several disease groups so they can be compared without overlap. |
| Density Plot | Smooth curve showing the spread of recovery time for one patient group. |
| Violin Plot | Combines density shape with median/quartiles to compare recovery across disease categories. |
| Histogram | Shows how patient age or visit count is distributed. |
| Boxplot | Compares median recovery/treatment time and outliers across groups. |

### Outputs:




- **Ridgeline plot:** Lets you compare many disease groups at once without the curves overlapping and becoming unreadable.
- **Density plot:** Shows the true shape of recovery times more smoothly than a histogram.
- **Violin plot:** Shows both the typical recovery rate and whether a group has two different patterns (fast vs slow recovery).
- **Histogram:** The simplest way to see how patient age or visits are spread across ranges.
- **Boxplot:** Quickly compares many disease categories and flags patients with unusually long recovery.

## How This Helps Healthcare Planning

- Improve treatment plans by comparing recovery patterns across groups.
- Spot patient subgroups that need a different treatment approach.

- Plan staff and beds based on patient age groups and visit frequency.
- Flag disease categories with unusually long recovery for further review.

### Rubrics for Calculation

| Criteria | Weightage | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning | Marks | |
|---|---|---|---|---|---|---|---|
| Understanding of Industry Problem | 20% | Comprehensive understanding of real-world problem and constraints | Good understanding with minor gaps in requirements | Basic understanding; some requirements unclear | Poor understanding; misinterprets problem context | | |
| Application of Engineering Concepts | 25% | Applies advanced and relevant concepts effectively to solve the problem | Applies appropriate concepts with minor errors | Limited application of concepts | Incorrect or no application of concepts | | |
| Solution Design | 25% | Innovative, practical, and feasible solution aligned with industry standards | Logical and feasible solution with minor gaps | Basic solution with limited feasibility | Unclear or impractical solution | | |
| Use of Tools | 15% | Effective use of modern tools, technologies, and real data | Appropriate use of tools with correct implementation | Limited or inconsistent use of tools | Minimal or incorrect use of tools | | |
| Analysis | 10% | Thorough analysis with validated results and performance evaluation | Good analysis with reasonable validation | Basic analysis with limited validation | Poor or no analysis | | |

| **Faculty In-charge** | **Course Coordinator** | **Head of Department** |
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| Signature: _______________ | Signature: _______________ | Signature: _______________ |
| Date: ___________________ | Date: ___________________ | Date: ___________________ |